



AI Usage Report

Course: CS202 — Object-Oriented Programming (APCS)
Project: Super Mario Bros. (C++17 / SFML 3.0.0 / Box2D 2.4.1)
Group: Group 04

1. Truong Quang Bao Duong (TV1 — Lead Architect & Integration)

- **Student ID:** 25125009
- **Role:** Team Leader, System Architect, Game State Machine, Level Integration, Factory & Observer Patterns, Box2D Physics Sync & Collision Dispatch, Camera Viewport & 2-Player Co-op / Versus Systems, Bowser Boss Battle & Bullet Bill Launcher, Parallax Background Renderer, Automated CTest Test Harness (38 Suites).

No.	Purpose / Technical Objective	AI Tool Used	Iterative Prompt Sequence
1	Cross-Platform CMake Build System & SFML 3.0.0 Integration	Antigravity IDE (Gemini 3.1 Pro)	Prompt 1: "How to write a cross-platform CMakeLists.txt in C++17 to link pre-packaged SFML 3.0.0 libraries for MinGW-w64 on Windows and Homebrew on macOS?"
2	Mermaid Class Diagram Generation for OOP Architecture	Antigravity IDE (Gemini 3.1 Pro)	Prompt 1: "Generate Mermaid class diagram syntax for a 2D Super Mario game showing inheritance between Entity, Character, and Mario classes." Prompt 2: "How to represent the Observer and EventBus design patterns cleanly using Mermaid syntax for our project documentation?"
3	Standardizing Coding Guidelines & File Structure	Antigravity IDE (Gemini 3.1 Pro)	Prompt 1: "Proofread and format our WEEKLY_PLAN.md and CODING_RULES.md into professional academic English with standard naming conventions and milestones."
4	Singleton EventBus Architecture (Observer Pattern)	Antigravity IDE (Gemini 3.1 Pro)	Prompt 1: "How to implement a robust EventBus in C++17 using a Singleton pattern to allow dynamic listener subscription and notification without iterator invalidation?" Prompt 2: "How to properly trigger sound events like PLAYER_JUMPED from the Physics module to the Audio module using a centralized EventBus without data payload dependencies?"
5	Factory Pattern (EntityFactory) Implementation	Antigravity IDE (Gemini 3.1 Pro)	Prompt 1: "How to implement a Factory pattern in C++17 to instantiate polymorphic game objects like Goomba and Coin using std::unique_ptr based on character codes from a level file?"
6	State Pattern Architecture (IGameState & GameManager)	Antigravity IDE (Gemini 3.1 Pro)	Prompt 1: "I noticed a crash when popping states inside event callbacks. Why does calling pushState/popState/changeState directly during update/processInput cause iterator invalidation or dangling pointer access?"
7	Box2D Physics Synchronization Order & Glitch Fixes	Antigravity IDE (Gemini 3.1 Pro)	Prompt 1: "/codex-workflow use heavy route. 'Please review the core physics and enemy logic.'" Prompt 2: "/codex-workflow use heavy route. 'Currently, there is a bug where enemies get stuck or float. Please investigate and fix it.'"

No.	Purpose / Technical Objective	AI Tool Used	Iterative Prompt Sequence
8	Hardware Keyboard Polling & Anti-Stuck Key Fix	Antigravity IDE (Gemini 3.7 Flash)	Prompt 1: "During rapid state changes, keys occasionally stay stuck in a held state. Implement physical hardware polling to detect and clear ghost keys."
9	Automated CTest Test Harness (38 Suites)	Antigravity IDE (Gemini 3.1 Pro)	Prompt 1: "Write independent assert-based CTest test suites to cover the core systems, and ensure they all pass before we push the release candidate to develop."
10	Reviewing PRs & Branch Integration Management	Antigravity IDE (Gemini 3.7 Flash)	Prompt 1: "Use the /review-pr command to check my latest changes on the develop branch." Prompt 2: "Review the feature/level-and-enemy branch. Select only the newest commits and ensure that TV4 is set as the sole author (do not add me as a co-author)."
11	Boss Bowser AI & Authentic Level 4 Architecture	Antigravity IDE (Gemini 3.1 Pro)	Prompt 1: "Can you search online and build World 1-4 for me using the available assets? Adjust the layout until it matches the reference image. If a specific tile is missing, search the asset files thoroughly. /goal" Prompt 2: "Please review Bowser's mechanics and animations. The fire breath looks a bit off right now. Look up references online to make it look authentic."
12	Directional Bullet Bill Launcher & Firebar Hazards	Antigravity IDE (Gemini 3.1 Pro)	Prompt 1: "Implement the rotating Firebar hazard orbiting center anchor blocks, and ensure the multi-height fire breath logic works cleanly for the Castle level."
13	Bullet Bill Artillery & Homing Logic	Antigravity IDE (Gemini 3.1 Pro)	Prompt 1: "Implement the Bullet Bill launcher. It needs to detect Mario's position and fire projectiles when he is not standing directly adjacent to the cannon."
14	Piranha Plant Pipe Emergence AI & Sensor Physics	Antigravity IDE (Gemini 3.1 Pro)	Prompt 1: "Add the pipe and the man-eating plant (Piranha Plant) into the second section. Configure its sensor physics so it stays hidden when Mario is standing near or on the pipe."

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15	Flying Cheep Cheep Enemies Implementation	Antigravity IDE (Gemini 3.7 Flash)	Prompt 1: "Read the files related to coordinates, physics, and camera. Create a detailed plan to implement the [Cheep Cheep] enemy and how I should spawn it, but do not modify the map yet." Prompt 2: "Now add the flying Cheep Cheeps (moving up and down) into Level 2, just before the elevator section."
16	Springboard (Trampoline) Physics & Jump Multiplier	Antigravity IDE (Gemini 3.1 Pro)	Prompt 1: "Add the springboard and horizontal elevator to provide alternative paths for crossing. Adjust the ceiling near Bowser so the elevator can move smoothly, and add the bridge release chain near the axe."
17	Moving Elevator Platform Kinematics	Antigravity IDE (Gemini 3.7 Flash)	Prompt 1: "The moving objects like the elevator need defined start and end points for their patrol paths. Also, adjust the speed so it's moderate and well-timed for platforming." Prompt 2: "Adjust the elevator by shifting it slightly to the right to create a better platforming rhythm."
18	Level 4 Hazard Placement & Rhythm Tuning	Antigravity IDE (Gemini 3.7 Flash)	Prompt 1: "Place the flying Cheep Cheeps so they jump slightly higher than the elevator. Tweak the placement of the first and second elevators so the jumping sequence looks beautiful and plays smoothly."
19	2-Player Co-op Viewport Math & Dual Midpoint Tracking	Antigravity IDE (Gemini 3.1 Pro)	Prompt 1: <code>"/codex-workflow use heavy route. 'In 2-player co-op mode, when players move to opposite edges of the screen, one character starts flickering. Also, in Level 3, players can swim left out-of-bounds at the start. Please investigate and fix.'"</code>
20	Co-op Invisible Ceiling Guard	Antigravity IDE (Gemini 3.7 Flash)	Prompt 1: "In co-op mode, if one player jumps to a high platform while the other stays below, the bottom player gets dragged off-screen. Implement a vertical separation limit."
21	Co-op Warp Pipes & Mario/Luigi Fire Animations Fix	Antigravity IDE (Gemini 3.7 Flash)	Prompt 1: "Why does Luigi trigger the 'giant' animation instead of turning into Fire Luigi when picking up a Fire Flower? Please analyze this logic bug carefully." Prompt 2: "Add a slight delay to the pipe transition animations. Some of them happen almost instantly which looks unnatural."

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22	2-Player Versus (PvP) Ceiling Clamp & Mechanics	Antigravity IDE (Gemini 3.1 Pro)	Prompt 1: "Implement a PvP ceiling clamp and a narrow patrol breakout AI to prevent players from escaping the arena bounds during versus mode."
23	Stage Select Flow & UI Design with Snapshots	Antigravity IDE (Gemini 3.7 Flash)	Prompt 1: "Try designing the Level Select interface using your image generation capabilities. The current assets are lacking a good background." Prompt 2: "Find some reference images online and fit them into the 4 stage squares. Remove the 'difficulty' and 'enemies' text to keep the UI clean. Keep adjusting until it looks perfect. /goal"
24	Multi-theme UI Adjustments & HUD Realignment	Antigravity IDE (Gemini 3.7 Flash)	Prompt 1: "On the main title screen, remove the extra text and 'Top' score display. Clearly state our team name as 'GROUP 04' and mention 'CS202'." Prompt 2: "Center the two warp pipes properly. Ensure all UI elements are aligned to the middle, and remove the period dot in the word 'Bros.'"
25	Dynamic Sub-Area Grid Theme Resolution	Antigravity IDE (Gemini 3.7 Flash)	Prompt 1: "In multi-theme levels like Level 3 and 4, there are some map geometries and palettes that look out of place. Please identify and fix the mismatched themes."

2. Nguyen Phan Minh Nhat (TV2)

- **Student ID:** 25125033
- **Role:** Engine, Game Loop, Texture Resource Management, AnimationSystem, Camera 4-Way Tracking & Deadzone, UI Menus & Integer Scaling Letterbox.

No.	Purpose / Technical Objective	AI Tool Used	Iterative Prompt Sequence
1	Suggest structure of the refactored Game class in a way that is friendly to future development.	Gemini 3.1 Pro	Prompt 1: "What are possible future hurdles in the developmental period that I may want to consider when I create class Game?" Prompt 2: "Please propose a skeleton structure for the class following the cautions we discussed."
2	Identify and add prefix "m_" to improperly named variables.	Gemini 3.1 Pro	Prompt 1: "Please identify all instances of 'window' and 'shape' in the code and add 'm_' prefixes to all of them."
3	Identify magic numbers and replace them with proper constants.	Gemini 3.1 Pro	Prompt 1: "Please find all instances of magic numbers and replace them with these unsigned int constants."
4	Suggest the way to merge the duplicate "private" label.	Gemini 3.1 Pro	Prompt 1: "What is the best way to merge the 2 duplicated 'private' labels and what are things I should look out for?"
5	Plan the structure and responsibilities of classes TextureManager, AnimationSystem, and Camera.	Gemini 3.1 Pro	Prompt 1: "Explain to me what each class TextureManager, AnimationSystem, and Camera should do and the methods that should be implemented." Prompt 2: "How do they work in tandem with each other?"
6	Point out invariants and bugs that may occur and suggest fixes.	Gemini 3.1 Pro	Prompt 1: "What are pitfalls that can happen when someone else uses these methods improperly and how do I ensure it crashes gracefully?" Prompt 2: "If deleting a texture from TextureManager that still has an active instance crashes the game and there's no way to strictly prevent that from inside the class, please update Doxygen to warn about it."

No.	Purpose / Technical Objective	AI Tool Used	Iterative Prompt Sequence
7	Implementation of classes TextureManager, AnimationSystem, and Camera.	Gemini 3.1 Pro	Prompt 1: "Please implement class TextureManager for me based on the features and method we discussed." Prompt 2: "Implement class AnimationSystem, like what we discussed." Prompt 3: "Use the same format we used, and based on the team's guideline, please implement class Camera."
8	Design a test main.cpp file to test the features.	Gemini 3.1 Pro	Prompt 1: "Design for me a main.cpp driver to test out the three classes, but most importantly to check TextureManager's ability to handle missing source and the clamping feature of Camera."
9	Propose and implement a hybrid between Y-axis locking and partial tracking of the character model's coordination for Camera.	Antigravity IDE (Gemini 3.1 Pro)	Prompt 1: "The Camera was implemented to strictly track the player model's coordination, which doesn't fit this game, propose a method to fix this." Prompt 2: "Help me implement the changes based on the plan we came up with."
10	Review the placeholder MenuState class and propose and implement the new functional structure.	Antigravity IDE (Gemini 3.1 Pro)	Prompt 1: "as for MenuState, before we do anything, please review the spritesheet hud.png and propose what we should do" Prompt 2: "Let's use this as reference. We will use the mushroom which is option B, but also, we should also use the number 0 and 1 from the text to fill in the squares below MARIO, WORLD, the gold counter, and the TOP score. No implementation yet, how would we do that?"

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11	Fine-tune the sprite-slicing components of the Menu.	Antigravity IDE (Gemini 3.1 Pro)	Prompt 1: "ok, the tilemap coordination is currently not correct but we will figure that out later. Let's deal with these 2 problems first, the position of every pixels are 1 pixel off to the left of the actual placeholder boxes, and please use the Nestopia scheme instead." Prompt 2: "ok, check out MenuState.cpp. The numbers are all correct now so keep them, but I don't want the mushroom to blink anymore so please remove that. Also, add the animation for the coin 16 pixels on the left of '00'. The coin animation has 6 frames, the next one 9 pixels to the last in the spritesheet. Please implement it for me and then let me figure out the number myself." Prompt 3: "Make it so that each frame takes twice as long." Prompt 4: "the demo is working well, however, except from the mushroom and the gold coin, everything else is dependent on save data? From all the magic numbers I have figured out, please rewrite MenuState.cpp so that it can accept input of these values and change accordingly, the 4 inputs that we are currently defaulting to."
12	Refactor the format of files to adhere to CODING_RULES.md .	Antigravity IDE (Gemini 3.1 Pro)	Prompt 1: "Now, please consult CODING_RULES.md again and check out the files we have changed in this session and fix them to ensure they follow the guideline." Prompt 2: "ok, I still need the header fixed, please look at how Camera.h is formatted and use similar Doxygen format accordingly."

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13	Implement a pilot version of letterbox and propose a better method.	Antigravity IDE (Gemini 3.1 Pro)	Prompt 1: "Help me implement a letterbox with integer scaling based on this <code>DisplayConfig</code>." Prompt 2: "Explain to me how the letterbox is being constructed right now. It's not working and the scaling is wrong, there is black space, but the part where it's not black also follows wrong scaling." Prompt 3: "Please implement it so that you lock the window size ratio to that of 1280x720, and don't floor the tracking. We lock the ratio so that when it scales, we only need to scale up the screen without having to do anything with the offset."
14	Implement new Camera deadzone logic that better fits the game.	Antigravity IDE (Gemini 3.1 Pro)	Prompt 1: "The default position of the Y-axis should be 2 tiles below where Mario spawns, but when Mario jumps higher than that, expand it so that the camera only goes at most a fix amount on top of Mario, and when he goes back to the deadzone the camera goes back to default position." Prompt 2: "Treat the bottom offset like the default position, when Mario falls down, the camera keeps following him with a 40 pixel buffer until it snaps back to the bottom offset"
15	Comprehend files made by another member and design plan to fix scaling issues of screens.	Antigravity IDE (Gemini 3.1 Pro)	Prompt 1: "I want you to investigate the pause screen and the game over screen. They seem to not be compatible for use with <code>DisplayConfig</code> yet as they are way too big, give me a plan to tackle it." Prompt 2: "PauseState and Winstate look good, however GameOverState is still way too big and should be cut down even more."

No.	Purpose / Technical Objective	AI Tool Used	Iterative Prompt Sequence
16	Design and implement the new system for screens.	Antigravity IDE (Gemini 3.1 Pro)	Prompt 1: "I want all the screens to used a shared UI layout helper. Please propose a design pattern for this. What are the changes, make sure to make them such that the subsequent tasks are easy to implement." Prompt 2: "Based on the anchors, implement Resume and Quit to Menu feature for <code>PauseState</code>." Prompt 3: "Implement Final Score indicator as well as Retry and Menu options for <code>GameOverState</code>." Prompt 4: "Implement Final Score indicator and Return to Menu option for <code>WinState</code>."
17	Check and revise the result based on criterias listed in <code>CODING_RULES</code> .	Antigravity IDE (Gemini 3.1 Pro)	Prompt 1: "please read <code>CODING_RULES.md</code> again and make sure the files we have altered adhere to the coding rules, apply doxygen format to header files, and especially turn magic numbers into constants and give instructions on what each of them does."

3. Nguyen Quoc Bao (TV3)

- **Student ID:** 25125079
- **Role:** Mario & Luigi Character Controller, Box2D Physics (PhysicsEngine), Mario Power-Up State Machine, CollisionManager & FireBall Projectiles.

No.	Purpose / Technical Objective	AI Tool Used	Iterative Prompt Sequence
1	Entity Folder Structure & Architecture Outline	Gemini 3.1 Pro	Prompt 1: "Create an outline for the entity folder structure in a 2D C++ game project."
2	Entity Class Inheritance Hierarchy Visualization	Gemini 3.1 Pro	Prompt 1: "Create a class diagram to visualize the relationship and inheritance between Entity, Character, and Mario classes."
3	Entity, Character, and Mario Initial Header Skeletons	Gemini 3.1 Pro	Prompt 1: "Implement C++ header code for Entity, Character, and Mario classes, making sure it strictly follows OOP concepts and project rules."
4	Box2D Physics Engine Architecture & Integration	Gemini 3.1 Pro	Prompt 1: "I want to use Box2D library for my physics mechanic implementation, make a plan to do it" Prompt 2: "Implement the plan step by step, make sure to follow the Coding rules and the OOP concepts."
5	Mario Physics Refinement & Collision Mechanics Plan	Gemini 3.1 Pro	Prompt 1: "Could you check my code and make a to-do list to complete my task"
6	Terminal Fall Velocity, Zero-Friction Wall Fix & Dynamic Hitbox Resizing	Gemini 3.1 Pro	Prompt 1: "Can you follow the to-do list and make changes to my code step-by-step" Prompt 2: "Make sure to prevent any hardcode and the code should be prepare for further update and debugging friendly "
7	Compiler Diagnostics, EventBus API Alignment & Clean Build Verification	Gemini 3.1 Pro	Prompt 1: "List out how the character should behave after i make the changes this week" Prompt 2: "now compile the code for debugging purpose"
8	CollisionManager Pointer Resolution, FireBall Entity & Mario State/Lives	Gemini 3.1 Pro	Prompt 1: "Implement the steps step by step according to the plan, make sure to review the checklist" Prompt 2: "the current main have nothing when compiled, rewrite the TV3Demo to run test my current code..."
9	Mid-Air Wall-Sticking & Zero Contact Friction Diagnosis	Gemini 3.1 Pro	Prompt 1: "i observed a bug, when i make contact with the wall mid air and i hold d, the gravity does not apply on the object anymore. Explain what is the problem and how to fix it"
10	Authentic NES Movement Physics, Skid Friction & FireBall Bouncing	Gemini 3.1 Pro	Prompt 1: "Make a plan to apply these physics and make sure it is Mario-like physics: Tune the moving mechanics for Mario... Make the fireball bounce"

No.	Purpose / Technical Objective	AI Tool Used	Iterative Prompt Sequence
11	Box2D Synchronous Body Deletion Crash Fix & Overhead Block Test Fixture	Gemini 3.1 Pro	Prompt 1: "There is a bug when the fire ball spawned and bounce , it crash the programme. Discover and fix the bug" Prompt 2: "give me a overhead block to test my jumping and collision with a block above" Prompt 3: "i do not recieve the debugging messages, fix that for me"
12	Code Quality & Zero-Hardcoding Compliance Audit	Gemini 3.1 Pro	Prompt 1: "Can you check that if the recent work is flexible and easy to expand and do not use any hardcode"
13	Merge Alignment Preparation & Cross-Module Collision Wrappers	Gemini 3.1 Pro	Prompt 1: "Read the review tv3 file and summary what have to be changed when merge with develop and further work. Make a to-do list out of it"
14	Remote Develop Workspace Synchronization	Gemini 3.1 Pro	Prompt 1: "pull from the newest develop branch"
15	Sprint 5 Error Report Remediation Analysis	Gemini 3.1 Pro	Prompt 1: "Make sure check TV3README.md for rules, remember that we are TV3, read implementation_plan_sprint5_error.md to make a plan to implement the fixes for the errors mentioned. Double check if the suggested plan is efficient"
16	Sprint 6 Refactoring & State Machine Optimization Roadmap	Gemini 3.1 Pro	Prompt 1: "Read the TV3README.md and make a plan to implement the Sprint 6 Refactoring & Optimization Roadmap (TV3), evaluating the to-do list and suggest for any improvement, make sure to follow the rules."
17	Branch Merging & Build Conflict Resolution	Gemini 3.1 Pro	Prompt 1: "Try to merge the current branch with the branch 'feature/level-and-enemy' branch"
18	Shift-Key Sprint Separation Mechanics & Verification Tests	Gemini 3.1 Pro	Prompt 1: "Is the walking and sprinting separated by shift, make a plan to implement the walking seperately"
19	120ms Jump Buffering, Held-Jump Stomp Boost & Co-op Head-Bounce	Gemini 3.1 Pro	Prompt 1: "Implement 120ms jump buffering in Mario.cpp so jump inputs pressed right before landing are not dropped." Prompt 2: "Ensure holding jump button during enemy stomp applies full jump force velocity boost matching NES canon." Prompt 3: "Update CollisionManager to support co-op partner head-bounce boost jumping without causing friendly fire."

No.	Purpose / Technical Objective	AI Tool Used	Iterative Prompt Sequence
20	4-Level Campaign Playthrough Physics Audit & Test Automation	Gemini 3.1 Pro	Prompt 1: "Verify Mario underwater swimming physics, buoyancy damping, and Castle hazard collision layers across release levels." Prompt 2: "Run full CTest test suite and document test results for TV3 deliverables in change_in_develop.md."

4. Tran Hong Vy (TV4)

- **Student ID:** 25125049
- **Role:** Level Design (World 1-1 -> 1-4, PvP Arena), TileMap Parser, Enemy AI Hierarchy (11 Enemy Species), Viewport Activation, SaveManager.

No.	Purpose / Technical Objective	AI Tool Used	Iterative Prompt Sequence
1	ASCII Level File Format & Initial Level Structure	ChatGPT (GPT-5.5 Sol)	Prompt 1: "Suggest me some format of game level."
2	Level & Enemy Module Validation and Development Planning	ChatGPT (GPT-5.5 Sol)	Prompt 1: "Here is my latest version folder in my branch feature/level-and-enemy. Check whether there are any logic errors, memory leaks, or violations of the team coding rules."
3	TileMap Loading, Validation, Queries & Rendering Helpers	ChatGPT (GPT-5.6 Thinking)	Prompt 1: "Help me improve TileMap::loadFromFile() , implement tile query and rendering helpers" Prompt 2: "Check TileMap.cpp for duplicated constant, outdated tile indices, invalid symbols, incorrect solid-tile rules, and verify that each level symbol maps to the correct tile in the 320x32 tileset."
4	Enemy Base Class & Goomba Patrol Implementation	ChatGPT (GPT-5.6 Thinking)	Prompt 1: "Help me implement Enemy and Goomba with horizontal patrol, stomp behaviour, public onWallCollision() , private reverseDirection() , and correct Box2D synchronization."
5	Goomba Animation, Stomp State & Safe Delayed Removal	ChatGPT (GPT-5.6 Thinking)	Prompt 1: "Help me implemen Goomba walking animation, the squished state, and delayed removal after 0.5 seconds while keeping the Box2D body handling safe." Prompt 2: "Double-check my Goomba.cpp implementation for incorrect includes, animation API misuse, trailing whitespace, outdated comments, and memory-safety issues"
6	Koopa State-Based Enemy AI & Shell Mechanics	ChatGPT (GPT-5.6 Thinking)	Prompt 1: "Help me create Koopa.h and Koopa.cpp with pattol behaviour and the states WALKING , SHELL_IDLE , and SHELL_SLIDING , using named constants instead of magic numbers." Prompt 2: "Help me add Koopa walking and shell animations and implement the state transition from walking to stationary state, stationary shell to sliding shell when kicked, and sliding shell back to stationary shell when stomped."

No.	Purpose / Technical Objective	AI Tool Used	Iterative Prompt Sequence
7	Level 1 Validation & Level 2 Gameplay Design	ChatGPT (GPT-5.6 Thinking)	Prompt 1: "Review level1.txt for valid symbols, consistent row widths, enemy spawn points, pipes, gaps, stone blocks, and finish-pole placement." Prompt 2: "Help me shortent and rebalance my Level 2 design into a 128-tile underground cave layout while preserving the original platform, enemy, pipe, coin, and stone ideas and satisfying the weekly-plan requirement that Level 2 is harder and contains few4 coins."
8	Consolidated Tileset Migration & TileFrame Mapping	ChatGPT (GPT-5.6 Sol)	Prompt 1: "Help me migrate TileMap from the old separate block textures to assets/textures/tiles/tileset.png while keeping the world tile size at 32x32 and using 16x16 source rectangles from the atlas." Prompt 2: "Check my TileFrames.h and TileFrames.cpp coordinates against the consolidated tileset and make sure ground, brick, used block, question block, pipe, flag, and pole use the correct source rectangles."
9	Level 1 Tutorial Flow & Gameplay Balance	ChatGPT (GPT-5.6 Sol)	Prompt 1: "Review Level 1 and verify that its tutorial flow introduces movement, coins, QuestionBlock, Mushroom, enemies, and the finish flag in a reasonable order without requiring unfair jumps."
10	Viewport-Based Enemy Activation & Cleanup	ChatGPT (GPT-5.6 Sol)	Prompt 1: "Help me change enemy behaviour so an enemies activates only when it's within 64 pixels in front of the viewport, and once activated it continues updating instead of freezing when it goes-off screen." Prompt 2: "How should i remove an activated enemy only after it is more than one full viewport behind the camera without resetting its state"
11	Koopa Ledge Detection & Enemy Atlas Migration	ChatGPT (GPT-5.6 Sol)	Prompt 1: "Add ledge detection to walking Koopa using TileMap so it reverae direction before walking off a platform, while allowing SHELL_SLIDING to fall into pits." Prompt 2: "The old individual Goomba and Koopa textures were removed. Help me migrate both enemies to assets/textures/enemies/enemies.png using the coordinayes documented in enemies_coordinate.md ."

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12	Koopa Shell Collision & State Transition Corrections	ChatGPT (GPT-5.6 Sol)	Prompt 1: "Review CollisionManager so the first Mario stomp changes Koopa only from WALKING to SHELL_IDLE and cannot also kick the shell during the same collision." Prompt 2: "Implement the SHELL_IDLE to SHELL_SLIDING transition so a lateral Mario collision kicks the shell away from Mario , while stoping a moving shell return it to SHELL_IDLE ."
13	Sliding Shell Enemy Defeat & Duplicate Collision Guard	ChatGPT (GPT-5.6 Sol)	Prompt 1: "Review the enemy-enemy collision logic so a sliding Koopa shell defeats another enemy exactly once, even if Box2D reports multiple contacts before the victim is removed."
14	Versioned SaveMagager Architecture & Persistence	ChatGPT (GPT-5.6 Sol)	Prompt 1: "Design SaveManager.h / .cpp following our coding rules, with SaveData version 1 storing highScore , highestUnlockedLevel , soundVolume , and musicVolume , and support an injected save path for tests." Prompt 2: "Implement loading and saving so high score and unlocked level only increase, audio volume persist within 0-100, and invalid or missing save files safely fall back to default data."
15	Atomic Save-File Replacement & SaveManager Regression Tests	ChatGPT (GPT-5.6 Sol)	Prompt 1: "Replace direct save-file writes with a temporary-file and save-replace workflow so the previous save remains intact if writing fails." Prompt 2: "Write assert-based SaveManager regression tests for valid, missing, corrupted, wrong version, lower score, reload-persistence, and temporary-file cleanup cases, and register them with CTest."
16	Coding-Rule Compliance & Implementation Cleanup	ChatGPT (GPT-5.6 Thinking)	Prompt 1: "Check the TV4 files for incorrect comments, trailing whitespace, naming issues, and other violation with the team coding-rules."

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17	**Level 3 Multi-Area Theme Transition & Pipe Routing Debugging	ChatGPT (GPT-5.6 Sol)	Prompt 1: "In Level 1, the intended transition is Overworld -> Underground , but the theme does not change correctly after entering the pipe. Please help me trace and fixthe transition logic." Prompt 2: "Horizontal pipes should also change their apprearance according to the active level theme. Please help me integrate the theme-dependent rendering logic." Prompt 3: "At the beginning of Level 3, the active area is Overworld , but the horizontal pipe is rendered using the Underwater theme. Please help me identify why the wrong theme is being resolved."
18	Theme-Aware Water/Lava Rendering & Environmental Layering	ChatGPT (GPT-5.6 Sol)	Prompt 1: "Help me design a shared level marker for water and lava so the actual texture and behaviour can be resolved according to the current LevelTheme ." Prompt 2: "Help me organise the surface and body texture mappings for water and lava across the Overworld, Underground, Castle, and Underwater theme." Prompt 3: "I want Podoboo to be rendered before the water/lava layer so that it visually appears behind the liquid surface. Where should be rendering order be changed."
19	Castle Tile Rendering , Filebar Base & TileFrames Correction	ChatGPT (GPT-5.6 Sol)	Prompt 1: "I want a solid castle block rendered behind each Firebar to act as its visual base. Also, the tile mapped to the = symbol is incorrect. Please help me correct the mapping."
20	Level 4 Castle Progression, Hazards & Endling Design	ChatGPT (GPT-5.6 Sol)	Prompt 1: "Help me design Level 4 as a Castle stage inspire by the original SuperMarioBros. structure, using the systems and assets already available in the project, including Podoboo, Firebar, Bowset, the axe, and the bridge sequence." Prompt 2: "Level 1-3 use the normal flagpole ending, but Level 4 should end with Toad after the Bowser section. Help me adapt the level-completion flow for this different ending condition."

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21	Toad NPC Integration & Level Completion Contract	ChatGPT (GPT-5.6 Sol)	Prompt 1: "I already modified the Toad header, please review the header and related implementation."

5. Le Phi Truyen (TV5)

- **Student ID:** 25125068
- **Role:** Command Pattern (`InputHandler`), Hardware Input Synchronization & Anti-Ghosting, `SoundManager` Singleton & Audio Pool, Dynamic HUD , Item Hierarchy & `QuestionBlock` .

No.	Purpose / Technical Objective	AI Tool Used	Iterative Prompt Sequence
1	Audio System Architecture & SoundManager Singleton Skeleton	Antigravity IDE (Gemini 3.5 Flash & Claude Opus 4.6)	Prompt 1: "Write the <code>include/core/SoundManager.h</code> header file as a Singleton skeleton – declare all methods for sound effects and background music playback but do not implement them yet." Prompt 2: "How to ensure the SoundManager Singleton constructor is private, delete copy and move constructors, and provide a thread-safe static <code>getInstance()</code> accessor in C++17?"
2	Concrete Input Commands Implementation & SoundManager EventBus Integration	Antigravity IDE (Claude Opus 4.6)	Prompt 1: "Implement <code>JumpCommand</code>, <code>MoveLeftCommand</code>, <code>MoveRightCommand</code>, and <code>PauseCommand</code> concrete classes inheriting from <code> ICommand</code>, targeting Mario character intent methods." Prompt 2: "Inherit <code>SoundManager</code> from <code>IObserver</code> and implement <code>onNotify(EventType)</code> to automatically trigger sound effects when game events like <code>PLAYER_JUMPED</code> or <code>COIN_COLLECTED</code> occur."
3	BGM Looping, Sound Preloading & Sensor-Based Coin Entity	Antigravity IDE (Claude Opus 4.6)	Prompt 1: "Add preloading for all SFX (jump, coin, stomp, powerup, death) and BGM tracks into <code>SoundManager</code> constructor to prevent audio loading latency during gameplay." Prompt 2: "Implement <code>Coin.h/.cpp</code> using a <code>Box2D b2_staticBody</code> with sensor fixtures so coins detect Mario contact without physical obstruction, award score, and publish <code>COIN_COLLECTED</code> event."
4	Key Release Undo Mechanism & Game Loop Command Decoupling	Antigravity IDE (Claude Opus 4.6)	Prompt 1: "Add undo logic on key release in <code>InputHandler::handleInput()</code> so releasing movement keys immediately halts Mario's horizontal acceleration." Prompt 2: "Integrate <code>InputHandler</code> as a member of <code>Game/PlayState</code>, replacing legacy <code>sf::Keyboard::isKeyPressed</code> polling in the main event loop."

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5	Power-Up Items Hierarchy (Mushroom, FireFlower, Star), Factory Registration & HUD	Antigravity IDE (Claude Opus 4.6)	Prompt 1: "Implement Mushroom (horizontal movement and Mario SUPER transformation), FireFlower (stationary FIRE transformation), and Star (bouncing kinematic entity with temporary invincibility timer)." Prompt 2: "Implement HUD class in include/ui/HUD.h and src/ui/HUD.cpp rendering player score, coin count, world name, and lives using sf::Text and NES pixel font with fallback loading."
6	Consolidated SpriteFrames Atlas Definition & Animation System Integration	ChatGPT (GPT-5.6 Sol)	Prompt 1: "Create include/core/SpriteFrames.h containing static sf::IntRect constants for Mario, Luigi, items, and UI spritesheets based on the consolidated NES texture atlases." Prompt 2: "Implement createManualAnimation() helper and auto-scale sprite adjustments during Box2D physics synchronization in Entity.cpp."
7	Dynamic QuestionBlock Entity, Coin Popup Kinematics & Tile Collision Stability	ChatGPT (GPT-5.6 Sol)	Prompt 1: "Implement QuestionBlock entity with hit bump animation, transitioning from active Question Block texture to Used Block texture upon head bump." Prompt 2: "Fix coin popup bounce physics and auto-scaling so collecting a coin spawns an animated rising coin before awarding score to HUD."
8	Advanced Input State Management & Extended Commands (Sprint, Shoot, Jump Release)	ChatGPT (GPT-5.6 Sol)	Prompt 1: "Add Released input trigger to InputState and support multiple key bindings per action in InputHandler to distinguish Pressed, Held, and Released states." Prompt 2: "Implement RunCommand (Shift key sprint acceleration), ShootCommand (X key FireBall shot request), and JumpReleaseCommand (variable jump height cutoff on button release)."

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9	Multi-Track Music Flow, 4-Voice Concurrent SFX Pool & Dynamic HUD Timer	ChatGPT (GPT-5.6 Sol)	Prompt 1: "Refactor SoundManager with MusicId enum supporting stage themes (Overworld, Underground, Castle), Star theme override/restoration, and dedicated Game Over / Win music." Prompt 2: "Add a 400-second level countdown timer to HUD with low-time warning SFX, one-shot timeout death trigger, and live MarioState label tracking (SMALL, SUPER, FIRE, STAR)."
10	Centralized ScoreRules Catalog, Collectible Guards & Automated Integration Tests	ChatGPT (GPT-5.6 Sol)	Prompt 1: "Create ScoreRules.h/.cpp catalog centralizing all game scoring values (Coin +100, Stomp +100, Power-Up +1000, Shell Kill +200) to serve as a single source of truth." Prompt 2: "Write automated regression test suite tests/InputStateTests.cpp and tests/TV5IntegrationTests.cpp covering simultaneous key presses, release timing, item collection cardinality, and audio persistence."
11	Input Binding Refinements & Deterministic Power-Up Body Tier Preservation	Antigravity IDE (Gemini 3.1 Pro)	Prompt 1: "How to cleanly preserve Mario body tier (Small Fire vs Super Fire) across FireFlower pickup and damage transitions without state desynchronization?" Prompt 2: "Add gameplay input suppression flags in PlayState so player controls are completely ignored during flagpole sequences and transformation animations."
12	Centralized Collision Dispatch & Cause-Specific Defeat SFX Cues	Antigravity IDE (Claude Opus 4.6)	Prompt 1: "Centralize enemy defeat into CollisionManager::defeatEnemy() so that stomp, shell kick, fireball hit, and star contact trigger dedicated event types and distinct sound cues." Prompt 2: "Add GAME_RESUMED audio handling to ensure SFX and music smoothly resume playback without channel clipping after unpausing."

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13	Multi-Theme SpriteFrames Catalogs & Dynamic LevelTheme Propagation	Antigravity IDE (Gemini 3.1 Pro)	Prompt 1: "Refactor SpriteFrames to support multi-theme catalogs (Overworld, Underground, Castle, Underwater) so tiles and QuestionBlocks automatically resolve theme-appropriate texture rectangles." Prompt 2: "Propagate LevelTheme through EntityFactory::createItem() and TileMap to ensure seamless rendering across themed sub-areas."
14	QuestionBlock Weighted Drop Probability & Audio Request Diagnostics	ChatGPT (GPT-5.6 Sol)	Prompt 1: "Implement a weighted random distribution for QuestionBlocks: 70% Coin, 15% Mushroom, and 15% FireFlower, ensuring deterministic results across playthroughs." Prompt 2: "Add diagnostic audio request counters in SoundManager to verify that each in-game action triggers exactly one SFX request without duplicate channel allocation."
15	Deterministic Item Handling, Audio Request Diagnostics & Damage Grace Pass-Through	ChatGPT (GPT-5.6 Sol)	Prompt 1: "Implement enemy pass-through and damage grace period for Mario after taking damage to prevent instant double hits." Prompt 2: "Write regression tests asserting that rejected FireBall requests or repeated collision contacts produce zero additional SFX events."
16	Character Selection Flow, Pipe Warp Transitions & Anti-Stuck Physics Tuning	Antigravity IDE (Gemini 3.7 Flash)	Prompt 1: "Implement character selection menu logic allowing players to toggle between Mario and Luigi, loading character-specific spritesheets dynamically." Prompt 2: "Tune movement acceleration, friction, and ceiling clearance collision routines to prevent Mario from getting stuck between adjacent solid blocks."
17	Stomp Combo Scoring System & Animated Score Popups	Antigravity IDE (Gemini 3.7 Flash)	Prompt 1: "Add progressive stomp combo scoring (100 -> 200 -> 400 -> 800 -> 1-Up) for consecutive enemy stomps without touching the ground, spawning animated score popups." Prompt 2: "Plan and execute release candidate verification gates ensuring all 37 CTest suites pass."

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18	2-Player Co-op & Versus PvP Modes Integration	Antigravity IDE (Gemini 3.7 Flash)	Prompt 1: "Design input dispatching for 2-Player Co-op mode, mapping Player 1 to WASD/Space and Player 2 to Arrow keys/Numpad with independent command execution." Prompt 2: "Implement PvP arena combat rules where stomping the other player triggers bounce displacement and life deduction within bounded camera confines."
19	OOP Architecture Modernization (EntityView, Creator Hierarchy & EventBus RAI)	Antigravity IDE (Claude Opus 4.6)	Prompt 1: "Refactor EntityFactory from a monolithic switch statement to a polymorphic Creator hierarchy (EnemyCreator, ItemCreator) adhering to the Open/Closed Principle." Prompt 2: "Implement RAII Subscription tokens for EventBus to guarantee automatic unsubscription upon observer destruction, eliminating dangling pointer risks." Prompt 3: "Create a read-only EntityView wrapper and Capability bitmask enum to allow safe read access to entity coordinates without exposing mutable state."
20	OOP Design Patterns Documentation, Final Asset Hygiene & Demo Script	ChatGPT (GPT-5.6 Sol)	Prompt 1: "Author comprehensive docs/design_patterns.md documenting Factory Method, Singleton, Observer, State, and Command patterns with class diagrams and C++17 implementations." Prompt 2: "Draft a complete 5-minute video demo narration script highlighting core mechanics, design patterns, co-op/PvP modes, and technical milestones."